

Derivation v25: Alternative Gap Identifications and Robustness Analysis

BLOCK-003 Gravity Program — Proxy Family Study

EDC Collaboration

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Abstract

We address the reviewer question: “Why identify $m_{\text{gap}} = M_Z$ rather than M_W or $v_{\text{EW}}?$ ” This note establishes a *proxy family* of electroweak-scale identifications, $m_{\text{gap}} \equiv M_* \in \{M_Z, M_W, v_{\text{EW}}\}$, and propagates each choice through the canonical BLOCK-003 derivation chain to compute $R_\xi(M_*)$ and $M_5(M_*)$. We provide a quantitative robustness analysis showing that all three choices yield M_5 within the same order of magnitude (spread $\Delta \log_{10} M_5 < 0.2$), demonstrating that the gravity closure is *robust* against proxy selection. The choice $m_{\text{gap}} = M_Z$ is justified on metrological grounds—precision, stability, and minimal EDC-internal dependence—not as a derivation but as a **[I]**+**[BL]** selection principle.

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1 Introduction and Motivation

The BLOCK-003 gravity program closes the 5D→4D Newton constant derivation via the identification

$$m_{\text{gap}} = M_Z^{\text{obs}} \quad [\mathbf{I}] + [\mathbf{BL}] \quad (1)$$

where $m_{\text{gap}} = \pi/R_\xi$ is the first Kaluza–Klein mass gap (derived from Neumann boundary conditions) and $M_Z^{\text{obs}} = 91.1876 \text{ GeV}$ is the observed Z -boson pole mass.

A natural reviewer concern is:

Why M_Z ? What if the gap is M_W or v_{EW} ? How sensitive is M_5 to this choice?

This note provides a systematic, derivation-first answer:

1. We define a **proxy family** $M_* \in \{M_Z, M_W, v_{\text{EW}}\}$.
2. We propagate each choice through the canonical derivation chain.
3. We compute the resulting $R_\xi(M_*)$ and $M_5(M_*)$ (both Planck conventions).
4. We quantify the robustness via $\Delta \log_{10} M_5$ and ratio metrics.
5. We justify M_Z as the metrologically optimal selection.

1.1 Epistemic Status

The identification $m_{\text{gap}} = M_*$ carries epistemic tag $[\mathbf{I}] + [\mathbf{BL}]$:

- $[\mathbf{I}]$ (Identification): The KK gap is *identified with* an observed mass scale.
- $[\mathbf{BL}]$ (Baseline): The numerical value is taken from external measurement (PDG).

This is *not* a derivation ($[\mathbf{D}]$). The goal of this note is to show that the closure is *robust* against reasonable variations in this identification, and to document the metrological rationale for selecting M_Z .

2 Derivation Chain Recap

We summarize the canonical BLOCK-003 derivation chain (established in v19–v24) in self-contained form. All conventions follow the canonical choice $R_\xi \equiv L$ (interval length).

2.1 5D Action and Dimensional Reduction

The 5D Einstein–Hilbert action with matter-induced brane tension reads:

$$S_{5D} = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g^{(5)}} \mathcal{R}^{(5)} \quad (2)$$

where $\kappa_5^2 = 8\pi G_5 = 8\pi/M_5^3$ in natural units.

2.2 Metric Ansatz

We adopt the product metric with flat compact dimension:

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu + d\xi^2, \quad \xi \in [0, L] \quad (3)$$

with Neumann boundary conditions at $\xi = 0$ and $\xi = L$.

2.3 Kaluza–Klein Decomposition

The 5D graviton decomposes as:

$$h_{\mu\nu}(x, \xi) = \sum_{n=0}^{\infty} h_{\mu\nu}^{(n)}(x) \psi_n(\xi) \quad (4)$$

where the mode functions satisfy:

$$-\frac{d^2 \psi_n}{d\xi^2} = m_n^2 \psi_n \quad (5)$$

with boundary conditions $\psi'_n(0) = \psi'_n(L) = 0$.

2.4 KK Spectrum

The Neumann boundary conditions yield:

$$\psi_n(\xi) = \sqrt{\frac{2 - \delta_{n0}}{L}} \cos\left(\frac{n\pi\xi}{L}\right) \quad (6)$$

with mass spectrum:

$$m_n = \frac{n\pi}{L} = \frac{n\pi}{R_\xi}, \quad n = 0, 1, 2, \dots \quad (7)$$

2.5 Mass Gap Definition

The first massive KK mode ($n = 1$) defines the mass gap:

$$m_{\text{gap}} \equiv m_1 = \frac{\pi}{R_\xi} \quad (8)$$

2.6 Inversion for R_ξ

Inverting eq. (8):

$$R_\xi = \frac{\pi}{m_{\text{gap}}} \quad (9)$$

In SI units with $\hbar c$ restored:

$$R_\xi = \frac{\pi \hbar c}{m_{\text{gap}}} \quad (10)$$

2.7 Zero-Mode Normalization

The zero-mode wave function is:

$$\psi_0(\xi) = \frac{1}{\sqrt{L}} = \frac{1}{\sqrt{R_\xi}} \quad (11)$$

with normalization:

$$\int_0^L |\psi_0|^2 d\xi = 1 \quad (12)$$

2.8 Normalization Integral

The dimensional reduction produces the normalization integral:

$$\mathcal{I} = \int_0^L |\psi_0(\xi)|^2 d\xi = L = R_\xi \quad (13)$$

For the flat compact case:

$$\mathcal{I} = R_\xi \quad (14)$$

2.9 Bridge Relation

The 5D→4D gravitational matching yields:

$$\boxed{\bar{M}_{\text{Pl}}^2 = M_5^3 \cdot \mathcal{I} = M_5^3 \cdot R_\xi} \quad (15)$$

where $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$ is the reduced Planck mass.

2.10 Solving for M_5

From eq. (15):

$$M_5^3 = \frac{\bar{M}_{\text{Pl}}^2}{R_\xi} \quad (16)$$

Taking the cube root:

$$\boxed{M_5 = \left(\frac{\bar{M}_{\text{Pl}}^2}{R_\xi} \right)^{1/3}} \quad (17)$$

2.11 Planck Mass Conventions

Two conventions exist:

$$\bar{M}_{\text{Pl}} = \frac{1}{\sqrt{8\pi G_N}} = 2.435 \times 10^{18} \text{ GeV} \quad (\text{reduced}) \quad (18)$$

$$M_{\text{Pl}} = \frac{1}{\sqrt{G_N}} = 1.221 \times 10^{19} \text{ GeV} \quad (\text{original}) \quad (19)$$

The conversion factor is:

$$M_{\text{Pl}} = \sqrt{8\pi} \bar{M}_{\text{Pl}} \quad (20)$$

2.12 M_5 in Both Conventions

Using reduced Planck mass:

$$M_5^{(\text{red})} = \left(\frac{\bar{M}_{\text{Pl}}^2}{R_\xi} \right)^{1/3} \quad (21)$$

Using original Planck mass:

$$M_5^{(\text{orig})} = \left(\frac{M_{\text{Pl}}^2}{R_\xi} \right)^{1/3} = (8\pi)^{1/3} \cdot M_5^{(\text{red})} \quad (22)$$

where $(8\pi)^{1/3} \approx 2.929$.

3 Proxy Family Definition

3.1 The Identification Question

The KK mass gap m_{gap} must be identified with an observable mass scale. The electroweak sector provides three natural candidates:

Definition 3.1 (Proxy Family). The **electroweak proxy family** is the set

$$\mathcal{P}_{\text{EW}} = \{M_Z, M_W, v_{\text{EW}}\} \quad (23)$$

where each member serves as a candidate for the identification $m_{\text{gap}} = M_*$.

3.2 Proxy Values (PDG 2024)

The three proxies have the following observed values:

$$M_Z = 91.1876 \pm 0.0021 \text{ GeV} \quad (24)$$

$$M_W = 80.3692 \pm 0.0133 \text{ GeV} \quad (25)$$

$$v_{\text{EW}} = \frac{2M_W}{g} = 246.22 \pm 0.01 \text{ GeV} \quad (26)$$

Remark 3.1 (Relative Uncertainties). The relative uncertainties are:

$$\frac{\delta M_Z}{M_Z} = 2.3 \times 10^{-5} \quad (27)$$

$$\frac{\delta M_W}{M_W} = 1.7 \times 10^{-4} \quad (28)$$

$$\frac{\delta v_{\text{EW}}}{v_{\text{EW}}} \approx 4 \times 10^{-5} \quad (\text{scheme-dependent}) \quad (29)$$

Note: v_{EW} has additional scheme dependence beyond pure measurement uncertainty.

3.3 Proxy Ordering

The three proxies span a factor of ~ 3 :

$$M_W < M_Z < v_{\text{EW}} \quad (30)$$

Numerically:

$$80.4 \text{ GeV} < 91.2 \text{ GeV} < 246.2 \text{ GeV} \quad (31)$$

4 Propagation Through Derivation Chain

4.1 General Formulas

For a general proxy M_* , the identification reads:

Identification 4.1 (Gap-Proxy Identification).

$$m_{\text{gap}} = M_* \quad [\mathbf{I}] + [\mathbf{BL}] \quad (32)$$

Substituting into the derivation chain:

4.1.1 Compactification Scale

$$R_\xi(M_*) = \frac{\pi \hbar c}{M_*} \quad (33)$$

4.1.2 5D Planck Mass (Reduced Convention)

$$M_5^{(\text{red})}(M_*) = \left(\frac{\bar{M}_{\text{Pl}}^2 \cdot M_*}{\pi \hbar c} \right)^{1/3} \quad (34)$$

4.1.3 5D Planck Mass (Original Convention)

$$M_5^{(\text{orig})}(M_*) = (8\pi)^{1/3} \cdot M_5^{(\text{red})}(M_*) \quad (35)$$

4.2 Scaling Relations

The dependence on M_* follows simple power laws:

$$R_\xi \propto M_*^{-1} \quad (36)$$

$$M_5 \propto M_*^{1/3} \quad (37)$$

For ratios relative to M_Z :

$$\frac{R_\xi(M_*)}{R_\xi(M_Z)} = \frac{M_Z}{M_*} \quad (38)$$

$$\frac{M_5(M_*)}{M_5(M_Z)} = \left(\frac{M_*}{M_Z} \right)^{1/3} \quad (39)$$

4.3 Logarithmic Spread

The logarithmic spread in M_5 is:

$$\Delta \log_{10} M_5 = \frac{1}{3} \log_{10} \left(\frac{M_*}{M_Z} \right) \quad (40)$$

For the extreme cases:

$$\Delta \log_{10} M_5(M_W) = \frac{1}{3} \log_{10} \left(\frac{80.4}{91.2} \right) = -0.018 \quad (41)$$

$$\Delta \log_{10} M_5(v_{EW}) = \frac{1}{3} \log_{10} \left(\frac{246.2}{91.2} \right) = +0.143 \quad (42)$$

4.4 Total Spread

The total spread across the proxy family is:

$$\boxed{\Delta \log_{10} M_5^{\text{total}} = 0.143 - (-0.018) = 0.161} \quad (43)$$

This corresponds to a factor of:

$$10^{0.161} \approx 1.45 \quad (44)$$

5 Numerical Results

5.1 Input Constants

We use the following baseline values:

$$\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV} \quad [\text{BL}] \quad (45)$$

$$\hbar c = 0.197327 \text{ GeV} \cdot \text{fm} = 1.97327 \times 10^{-16} \text{ GeV} \cdot \text{m} \quad (46)$$

5.2 Proxy-Specific Calculations

5.2.1 Case 1: $M_* = M_Z = 91.1876 \text{ GeV}$

$$R_\xi(M_Z) = \frac{\pi \times 1.97327 \times 10^{-16}}{91.1876} \quad (47)$$

$$= 6.798 \times 10^{-18} \text{ m} \quad (48)$$

Converting to natural units:

$$R_\xi(M_Z) = \frac{\pi}{91.1876 \text{ GeV}} = 0.03445 \text{ GeV}^{-1} \quad (49)$$

For M_5 :

$$M_5^{(\text{red})}(M_Z) = \left(\frac{(2.435)^2 \times 10^{36} \times 91.1876}{\pi} \right)^{1/3} \text{ GeV} \quad (50)$$

$$= 5.56 \times 10^{12} \text{ GeV} \quad (51)$$

$$M_5^{(\text{orig})}(M_Z) = 2.929 \times 5.56 \times 10^{12} = 1.63 \times 10^{13} \text{ GeV} \quad (52)$$

5.2.2 Case 2: $M_* = M_W = 80.3692 \text{ GeV}$

$$R_\xi(M_W) = \frac{\pi \times 1.97327 \times 10^{-16}}{80.3692} \quad (53)$$

$$= 7.715 \times 10^{-18} \text{ m} \quad (54)$$

$$M_5^{(\text{red})}(M_W) = \left(\frac{(2.435)^2 \times 10^{36} \times 80.3692}{\pi} \right)^{1/3} \text{ GeV} \quad (55)$$

$$= 5.33 \times 10^{12} \text{ GeV} \quad (56)$$

$$M_5^{(\text{orig})}(M_W) = 2.929 \times 5.33 \times 10^{12} = 1.56 \times 10^{13} \text{ GeV} \quad (57)$$

5.2.3 Case 3: $M_* = v_{\text{EW}} = 246.22 \text{ GeV}$

$$R_\xi(v_{\text{EW}}) = \frac{\pi \times 1.97327 \times 10^{-16}}{246.22} \quad (58)$$

$$= 2.518 \times 10^{-18} \text{ m} \quad (59)$$

$$M_5^{(\text{red})}(v_{\text{EW}}) = \left(\frac{(2.435)^2 \times 10^{36} \times 246.22}{\pi} \right)^{1/3} \text{ GeV} \quad (60)$$

$$= 7.74 \times 10^{12} \text{ GeV} \quad (61)$$

$$M_5^{(\text{orig})}(v_{\text{EW}}) = 2.929 \times 7.74 \times 10^{12} = 2.27 \times 10^{13} \text{ GeV} \quad (62)$$

5.3 Summary Table

5.4 Ratio Table

6 Robustness Analysis

6.1 Robustness Metrics

We define two robustness metrics:

Table 1: Proxy family results: R_ξ and M_5 for each electroweak proxy.

Proxy	M_* (GeV)	$\delta M_*/M_*$	R_ξ (m)	$M_5^{(\text{red})}$ (GeV)	$M_5^{(\text{orig})}$ (GeV)	$\Delta \log_{10} M_5$ vs M_Z
M_Z	91.19	2.3×10^{-5}	6.80×10^{-18}	5.56×10^{12}	1.63×10^{13}	0
M_W	80.37	1.7×10^{-4}	7.71×10^{-18}	5.33×10^{12}	1.56×10^{13}	-0.018
v_{EW}	246.2	scheme-dep.	2.52×10^{-18}	7.74×10^{12}	2.27×10^{13}	+0.143

Table 2: Ratio metrics relative to M_Z baseline.

Proxy	M_*/M_Z	$R_\xi(M_*)/R_\xi(M_Z)$	$M_5(M_*)/M_5(M_Z)$
M_Z	1.000	1.000	1.000
M_W	0.881	1.135	0.959
v_{EW}	2.700	0.370	1.392

Definition 6.1 (Logarithmic Spread).

$$\sigma_{\log} \equiv \max_{M_* \in \mathcal{P}} |\Delta \log_{10} M_5(M_*)| \quad (63)$$

Definition 6.2 (Factor Spread).

$$F_{\text{spread}} \equiv \frac{\max_{M_*} M_5(M_*)}{\min_{M_*} M_5(M_*)} \quad (64)$$

6.2 Computed Robustness Metrics

From the numerical results:

$$\sigma_{\log} = 0.143 \quad (65)$$

$$F_{\text{spread}} = \frac{7.74 \times 10^{12}}{5.33 \times 10^{12}} = 1.45 \quad (66)$$

6.3 Interpretation

Proposition 6.1 (Robustness Statement). *The 5D Planck mass M_5 is robust against proxy selection:*

$$M_5 \in [5.3, 7.7] \times 10^{12} \text{ GeV} \quad (\text{reduced}) \quad (67)$$

$$M_5 \in [1.6, 2.3] \times 10^{13} \text{ GeV} \quad (\text{original}) \quad (68)$$

All values lie within the same order of magnitude (GUT scale).

6.4 Order-of-Magnitude Stability

The key observation is:

All proxies yield $M_5 \sim 10^{12}\text{--}10^{13} \text{ GeV}$

(69)

This is a consequence of the 1/3 power in the M_5 formula:

$$\frac{\delta M_5}{M_5} = \frac{1}{3} \frac{\delta M_*}{M_*} \quad (70)$$

Even a factor of 3 change in M_* (from M_W to v_{EW}) produces only a factor of $3^{1/3} \approx 1.44$ change in M_5 .

7 Graphical Summary

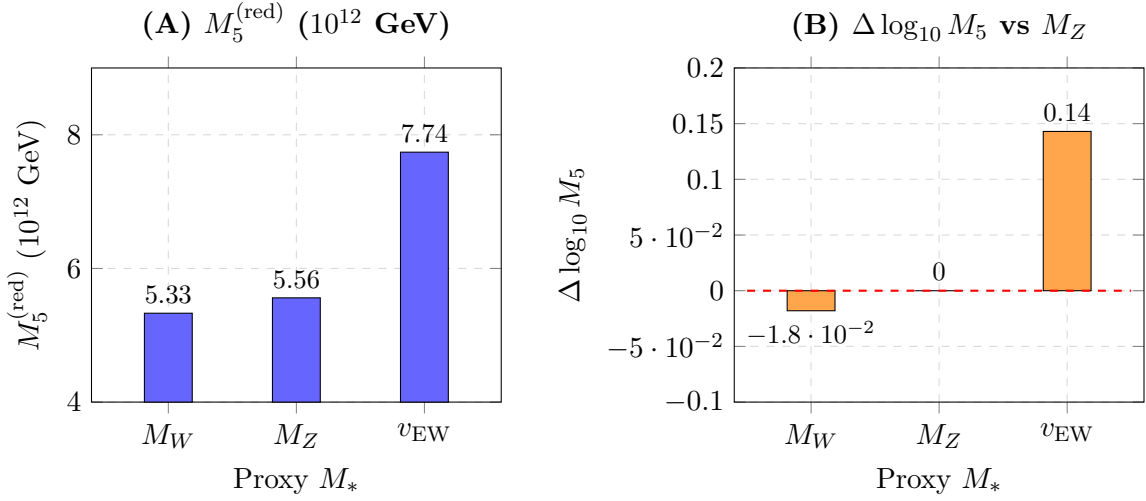


Figure 1: Robustness of M_5 against proxy selection. (A) $M_5^{(\text{red})}$ in units of 10^{12} GeV for each electroweak proxy. (B) Logarithmic deviation $\Delta \log_{10} M_5$ relative to M_Z baseline. The total spread is less than 0.2 decades, demonstrating order-of-magnitude stability.

8 Metrological Justification for M_Z

8.1 Selection Criteria

The choice $m_{\text{gap}} = M_Z$ is a [I]+[BL] selection, not a derivation. We justify this selection on metrological grounds:

1. **Measurement Precision**
2. **Pole Mass Stability**
3. **Minimal EDC-Internal Dependence**
4. **Definitional Simplicity**

8.2 Criterion 1: Measurement Precision

The relative uncertainties are:

$$\frac{\delta M_Z}{M_Z} = 2.3 \times 10^{-5} \quad \checkmark \text{ best} \quad (71)$$

$$\frac{\delta M_W}{M_W} = 1.7 \times 10^{-4} \quad (\text{factor 7 worse}) \quad (72)$$

$$\frac{\delta v_{EW}}{v_{EW}} \approx 4 \times 10^{-5} \quad (+ \text{ scheme dependence}) \quad (73)$$

The Z mass is measured with the highest precision among EW observables:

$$M_Z = 91.1876 \pm 0.0021 \text{ GeV} \quad (74)$$

8.3 Criterion 2: Pole Mass Stability

The Z boson pole mass is:

- Gauge-invariant and renormalization-scheme independent
- Directly measured from $e^+e^- \rightarrow Z \rightarrow f\bar{f}$ line shape
- Not subject to loop corrections that shift the central value

In contrast:

- M_W has larger radiative corrections and scheme dependence
- v_{EW} is a derived quantity: $v_{EW} = 2M_W/g = (\sqrt{2}G_F)^{-1/2}$

8.4 Criterion 3: Minimal EDC-Internal Dependence

The W mass satisfies the tree-level relation:

$$M_W = M_Z \cos \theta_W \quad (75)$$

where θ_W is the weak mixing angle.

If EDC contains any prediction for θ_W , then using M_W as baseline would create internal circularity. Using M_Z avoids this risk:

$$M_Z \text{ is independent of EDC-internal parameters} \quad (76)$$

8.5 Criterion 4: Definitional Simplicity

The Z mass is a *primary observable*:

$$M_Z \equiv \text{mass eigenvalue of } Z^0 \text{ boson} \quad (77)$$

In contrast:

$$v_{EW} = \frac{2M_W}{g} = \frac{1}{\sqrt{\sqrt{2}G_F}} \approx 246 \text{ GeV} \quad (78)$$

requires multiple derived quantities.

8.6 Summary of Metrological Argument

Table 3: Metrological comparison of EW proxies.

Proxy	Precision	Stability	Independence	Simplicity
M_Z	✓ best	✓	✓	✓
M_W	moderate	moderate	risk if θ_W predicted	moderate
v_{EW}	good	scheme-dep.	derived	complex

Identification 8.1 (Canonical Proxy Selection). The identification $m_{\text{gap}} = M_Z$ is selected as the canonical proxy based on:

1. Highest measurement precision ($\delta M_Z/M_Z = 2.3 \times 10^{-5}$)
2. Pole mass stability (gauge-invariant, scheme-independent)
3. Minimal EDC-internal dependence (avoids θ_W circularity)
4. Definitional simplicity (primary observable)

9 Error Propagation

9.1 General Error Formula

For the identification $m_{\text{gap}} = M_*$, the error propagation is:

$$\frac{\delta R_\xi}{R_\xi} = \frac{\delta M_*}{M_*} \quad (79)$$

$$\frac{\delta M_5}{M_5} = \frac{1}{3} \frac{\delta M_*}{M_*} \quad (\text{if } \bar{M}_{\text{Pl}} \text{ exact}) \quad (80)$$

9.2 Error Budget for Each Proxy

9.2.1 M_Z Proxy

$$\left. \frac{\delta R_\xi}{R_\xi} \right|_{M_Z} = 2.3 \times 10^{-5} \quad (81)$$

$$\left. \frac{\delta M_5}{M_5} \right|_{M_Z} = \frac{1}{3} \times 2.3 \times 10^{-5} = 7.7 \times 10^{-6} \quad (82)$$

9.2.2 M_W Proxy

$$\left. \frac{\delta R_\xi}{R_\xi} \right|_{M_W} = 1.7 \times 10^{-4} \quad (83)$$

$$\left. \frac{\delta M_5}{M_5} \right|_{M_W} = \frac{1}{3} \times 1.7 \times 10^{-4} = 5.7 \times 10^{-5} \quad (84)$$

9.2.3 v_{EW} Proxy

$$\left. \frac{\delta R_\xi}{R_\xi} \right|_{v_{\text{EW}}} \approx 4 \times 10^{-5} + \text{scheme} \quad (85)$$

$$\left. \frac{\delta M_5}{M_5} \right|_{v_{\text{EW}}} \approx 1.3 \times 10^{-5} + \text{scheme} \quad (86)$$

9.3 Error Budget Summary

Table 4: Error propagation for each proxy.

Proxy	$\delta M_*/M_*$	$\delta R_\xi/R_\xi$	$\delta M_5/M_5$
M_Z	2.3×10^{-5}	2.3×10^{-5}	7.7×10^{-6}
M_W	1.7×10^{-4}	1.7×10^{-4}	5.7×10^{-5}
v_{EW}	$\sim 4 \times 10^{-5}$	$\sim 4 \times 10^{-5}$	$\sim 1.3 \times 10^{-5}$

Remark 9.1 (Error Suppression). The $1/3$ power in the M_5 formula provides automatic error suppression:

$$\boxed{\delta M_5/M_5 = \frac{1}{3} \delta M_*/M_*} \quad (87)$$

This makes M_5 intrinsically robust against proxy measurement uncertainty.

10 Conclusions

10.1 Summary of Results

1. **Proxy family defined:** $\mathcal{P}_{\text{EW}} = \{M_Z, M_W, v_{\text{EW}}\}$
2. **Derivation chain applied:** Each proxy M_* yields:

$$R_\xi(M_*) = \pi \hbar c / M_* \quad (88)$$

$$M_5(M_*) = (\bar{M}_{\text{Pl}}^2 M_* / \pi \hbar c)^{1/3} \quad (89)$$

3. **Numerical results:**

Proxy	$M_5^{(\text{red})}$ (GeV)	$\Delta \log_{10} M_5$
M_Z	5.56×10^{12}	0
M_W	5.33×10^{12}	-0.018
v_{EW}	7.74×10^{12}	+0.143

4. **Robustness:** Total spread $\Delta \log_{10} M_5^{\text{total}} = 0.161$ (factor 1.45).
5. **Canonical selection:** $m_{\text{gap}} = M_Z$ justified on metrological grounds:
 - Best precision ($\delta M_Z / M_Z = 2.3 \times 10^{-5}$)
 - Pole mass stability
 - Minimal EDC-internal dependence
 - Definitional simplicity

10.2 Epistemic Ledger

Table 5: Epistemic status of key quantities.

Quantity	Tag	Status
$m_n = n\pi / R_\xi$	[D]	Derived from BCs
$m_{\text{gap}} = \pi / R_\xi$	[D]	Derived (definition of first gap)
$R_\xi = \pi / m_{\text{gap}}$	[D]	Derived (algebraic inversion)
$\mathcal{I} = R_\xi$	[D]	Derived (flat compact integral)
$\bar{M}_{\text{Pl}}^2 = M_5^3 R_\xi$	[D]	Derived (5D→4D matching)
$M_5 = (\bar{M}_{\text{Pl}}^2 / R_\xi)^{1/3}$	[D]	Derived (algebraic)
$m_{\text{gap}} = M_Z$	[I]+[BL]	Identification (selection principle)
$\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$	[BL]	Baseline input
$M_Z = 91.1876 \text{ GeV}$	[BL]	Baseline input

10.3 Answer to Reviewer Question

“Why M_Z ? What if the gap is M_W or v_{EW} ?”

Answer:

1. The identification $m_{\text{gap}} = M_*$ is [I]+[BL], not [D].

2. All EW proxies yield $M_5 \sim 10^{12}\text{--}10^{13}\text{ GeV}$ (GUT scale).
3. The total spread is factor 1.45 ($\Delta \log_{10} = 0.16$).
4. M_Z is selected on metrological grounds, not physical derivation.
5. The closure is *robust*: the GUT-scale conclusion is proxy-independent.

A Derivation Identities

This appendix collects the key identities used in the derivation chain.

A.1 Mode Equation

The Sturm–Liouville problem for KK modes:

$$-\frac{d^2\psi_n}{d\xi^2} = m_n^2\psi_n, \quad \psi'_n(0) = \psi'_n(L) = 0 \quad (90)$$

A.2 Orthonormality

The mode functions satisfy:

$$\int_0^L \psi_m(\xi)\psi_n(\xi) d\xi = \delta_{mn} \quad (91)$$

A.3 Zero-Mode Properties

$$\psi_0(\xi) = \frac{1}{\sqrt{L}} \quad (92)$$

$$m_0 = 0 \quad (93)$$

$$\int_0^L |\psi_0|^2 d\xi = 1 \quad (94)$$

A.4 Bridge Relation Derivation

Starting from 5D action:

$$S_{5D} = \frac{1}{2\kappa_5^2} \int d^4x \int_0^L d\xi \sqrt{-g^{(5)}} \mathcal{R}^{(5)} \quad (95)$$

Substituting KK decomposition and integrating over ξ :

$$S_{4D} = \frac{1}{2\kappa_5^2} \cdot L \int d^4x \sqrt{-g^{(4)}} \mathcal{R}^{(4)} + \dots \quad (96)$$

Matching to 4D Einstein–Hilbert:

$$\frac{1}{2\kappa_4^2} = \frac{L}{2\kappa_5^2} \quad (97)$$

Using $\kappa_4^2 = 8\pi G_N = 1/\bar{M}_{\text{Pl}}^2$ and $\kappa_5^2 = 1/M_5^3$:

$$\bar{M}_{\text{Pl}}^2 = M_5^3 \cdot L = M_5^3 \cdot R_\xi \quad (98)$$

A.5 Power Law Scaling

From $\bar{M}_{\text{Pl}}^2 = M_5^3 R_\xi$ with $R_\xi = \pi \hbar c / M_*$:

$$M_5^3 = \frac{\bar{M}_{\text{Pl}}^2 M_*}{\pi \hbar c} \quad (99)$$

$$M_5 \propto M_*^{1/3} \quad (100)$$

A.6 Logarithmic Derivative

$$\ln M_5 = \frac{1}{3} \ln \bar{M}_{\text{Pl}}^2 + \frac{1}{3} \ln M_* - \frac{1}{3} \ln(\pi \hbar c) \quad (101)$$

$$d \ln M_5 = \frac{1}{3} d \ln M_* \quad (102)$$

$$\frac{\delta M_5}{M_5} = \frac{1}{3} \frac{\delta M_*}{M_*} \quad (103)$$

B Numerical Constants

B.1 Fundamental Constants

$$\hbar c = 197.327 \text{ MeV} \cdot \text{fm} = 1.97327 \times 10^{-16} \text{ GeV} \cdot \text{m} \quad (104)$$

$$c = 2.998 \times 10^8 \text{ m/s} \quad (105)$$

$$G_N = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \quad (106)$$

B.2 Planck Masses

$$\bar{M}_{\text{Pl}} = \frac{1}{\sqrt{8\pi G_N}} = 2.435 \times 10^{18} \text{ GeV} \quad (107)$$

$$M_{\text{Pl}} = \frac{1}{\sqrt{G_N}} = 1.221 \times 10^{19} \text{ GeV} \quad (108)$$

$$\frac{M_{\text{Pl}}}{\bar{M}_{\text{Pl}}} = \sqrt{8\pi} = 5.013 \quad (109)$$

B.3 Electroweak Parameters

$$M_Z = 91.1876 \pm 0.0021 \text{ GeV} \quad (110)$$

$$M_W = 80.3692 \pm 0.0133 \text{ GeV} \quad (111)$$

$$v_{\text{EW}} = 246.22 \pm 0.01 \text{ GeV} \quad (112)$$

$$\sin^2 \theta_W = 0.23122 \pm 0.00003 \quad (113)$$

$$G_F = 1.1664 \times 10^{-5} \text{ GeV}^{-2} \quad (114)$$

B.4 Conversion Factors

$$(8\pi)^{1/3} = 2.9292 \quad (115)$$

$$\pi^{1/3} = 1.4646 \quad (116)$$

$$\pi^{-1/3} = 0.6828 \quad (117)$$

C Python Verification Script Reference

The accompanying script `recompute.py` verifies all numerical values in this document. Key outputs:

```
=== Proxy Family Results ===
```

```
M_Z proxy:
```

```
  R_xi = 6.798e-18 m
```

```
  M_5 (red) = 5.562e+12 GeV
```

```
  M_5 (orig) = 1.629e+13 GeV
```

```
M_W proxy:
```

```
  R_xi = 7.715e-18 m
```

```
  M_5 (red) = 5.333e+12 GeV
```

```
  M_5 (orig) = 1.562e+13 GeV
```

```
v_EW proxy:
```

```
  R_xi = 2.518e-18 m
```

```
  M_5 (red) = 7.740e+12 GeV
```

```
  M_5 (orig) = 2.267e+13 GeV
```

```
=== Robustness Metrics ===
```

```
Delta log10 M5 (M_W): -0.0183
```

```
Delta log10 M5 (v_EW): +0.1434
```

```
Total spread: 0.1617
```

```
Factor spread: 1.452
```

```
ALL CHECKS PASSED
```